

PHANTOM PROTOCOL



*A Verifiable, Post-Quantum, GPA-Resistant
Anonymous Communication Protocol*

WHITEPAPER v0.1 — DRAFT

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ABSTRACT

Phantom Protocol is a novel anonymous communication protocol designed to withstand a Global Passive Adversary (GPA) — including nation-state surveillance infrastructure. Unlike existing systems such as Tor, I2P, Katzenpost, and Nym, Phantom Protocol combines five properties no single protocol has simultaneously achieved: (1) cryptographically verifiable mixing via zero-knowledge proofs, eliminating the need to trust relay nodes; (2) a fully trustless PKI with no directory authorities, based on DHT-native node discovery and zk-credential admission; (3) native hidden service support enabling anonymous server hosting; (4) true peer-to-peer architecture where every client is a routing node; and (5) a post-quantum cryptographic foundation deployed from inception. A dual-mode design balances usability with extreme anonymity guarantees, offering Sphinx-based mix routing for general use and a DC-Net broadcast mode for information-theoretically anonymous transmission. This document presents the protocol's threat model, architectural design, cryptographic primitives, packet format, and implementation roadmap.

1. Introduction

The modern internet was not designed with privacy as a fundamental property. Decades of commercial surveillance, state interception programmes, and mass data collection have created a network where anonymity is the exception, not the rule. Existing privacy tools — VPNs, Tor, I2P — represent meaningful progress but were designed to address a different, weaker threat model than the one that exists today.

The revelations of global surveillance programmes demonstrated that well-funded nation-state adversaries operate at a scale that defeats every widely deployed anonymity system. A Global Passive Adversary (GPA) capable of observing a significant fraction of internet traffic simultaneously can deanonymize Tor users through traffic correlation, can compromise VPN providers through legal compulsion, and can identify participants in pseudonymous systems through long-term statistical analysis.

Phantom Protocol is designed from the ground up to be the answer to that adversary. It does not seek to retrofit privacy onto the existing internet — it defines a new communication layer that makes anonymity a mathematical guarantee rather than an operational assumption.

1.1 Design Philosophy

Three principles guide every architectural decision in Phantom Protocol:

- ▶ **Trust nothing.** Trust nothing. Every node, every authority, every third party is a potential adversary. The protocol is correct even when every relay is malicious.
- ▶ **Prove everything.** Prove everything. Anonymity guarantees must be cryptographically verifiable, not assumed. Nodes must prove correct behaviour, not merely promise it.
- ▶ **No centre.** No centre. Any component that can be targeted — a directory server, a certificate authority, a foundation — is a vulnerability. The protocol has no such components.

1.2 Scope of This Document

This whitepaper specifies the threat model, high-level architecture, cryptographic primitives, packet format, routing mechanisms, node admission, and implementation roadmap for Phantom Protocol version 1.0. It is intended as the foundational reference for protocol implementors, security researchers, and contributors.

2. Threat Model

Phantom Protocol is designed to provide strong anonymity guarantees against the most powerful realistic adversary class: the Global Passive Adversary (GPA).

2.1 Adversary Capabilities

We assume an adversary with the following capabilities:

- ▶ Full traffic observation: the adversary can observe all packets on all links between all nodes simultaneously, including timing, volume, and packet sizes.
- ▶ Node compromise: the adversary may operate or compromise an arbitrary fraction of relay nodes, up to but not including a global majority.
- ▶ Legal compulsion: the adversary can compel any identifiable legal entity — including protocol developers, foundation members, or hosting providers — to cooperate or cease operations.
- ▶ Long-term storage: the adversary stores all observed traffic indefinitely for retrospective analysis.
- ▶ Computational power: the adversary possesses current and near-future classical and quantum computing capabilities, including cryptanalytically relevant quantum computers within a 10-year horizon.
- ▶ Active interference: the adversary may inject, delay, replay, or drop packets on any link they control.

2.2 Adversary Limitations

The adversary cannot:

- ▶ Break information-theoretically secure constructions (DC-Net mode).
- ▶ Compromise a majority of honest nodes simultaneously.
- ▶ Break post-quantum cryptographic primitives standardised by NIST (CRYSTALS-Kyber, CRYSTALS-Dilithium).
- ▶ Force a user's device to reveal keys without physical access.

2.3 Anonymity Goals

Phantom Protocol provides the following formal anonymity properties:

- ▶ Sender anonymity: an adversary observing all network traffic cannot identify which node originated a given message with probability greater than $1/k$, where k is the anonymity set size.
- ▶ Receiver anonymity: the destination of a message is not revealed to any relay node or passive observer.
- ▶ Relationship anonymity: the association between sender and receiver cannot be established through traffic analysis.
- ▶ Unlinkability: multiple messages from the same sender cannot be correlated as originating from a single entity.
- ▶ Forward secrecy: compromise of long-term keys does not compromise past session keys or past communications.
- ▶ Post-quantum security: security properties hold against adversaries with access to cryptanalytically relevant quantum computers.

⚠ Fundamental Tradeoff

Phantom Protocol acknowledges the Anonymity Trilemma: strong anonymity requires overhead in latency and bandwidth. The protocol is honest about this tradeoff. Extreme anonymity (DC-Net mode) incurs latency of seconds to minutes. Standard mode (Sphinx mix routing) incurs latency of 200–800ms. Neither mode is suitable for real-time voice or video. Users must make an informed choice.

3. Prior Work and Differentiation

Phantom Protocol builds on decades of academic and engineering work in anonymous communications. This section analyses the four most relevant prior systems and identifies the precise gaps that Phantom Protocol fills.

3.1 Tor

Tor uses onion routing with three-hop circuits and a centralised directory authority model. It provides low-latency anonymity suitable for web browsing but is explicitly not designed to resist a GPA. A network-level adversary observing entry and exit can perform end-to-end traffic correlation with high confidence. Directory authorities represent a legal and operational attack surface. Tor's threat model explicitly excludes the adversary class that Phantom Protocol targets.

3.2 I2P

I2P uses garlic routing — a variant of onion routing that bundles multiple messages — and a distributed DHT-based directory. It is more resistant than Tor to directory attacks but similarly vulnerable to traffic analysis by a GPA. I2P's internal network (eepsites) provides some hidden service functionality, but the system is unproven against the adversary class Phantom targets.

3.3 Katzenpost

Katzenpost is the most technically advanced existing mix network and the closest predecessor to Phantom Protocol. It uses Sphinx packets, stratified topology, and post-quantum cryptography. However, it has three fundamental limitations that Phantom Protocol addresses: its PKI relies on directory authorities that are a legal attack surface; its mixing is unverified (nodes are trusted to shuffle honestly with no cryptographic proof); and clients are not routing nodes, limiting the anonymity set.

3.4 Nym

Nym is a production mix network based on the Loopix architecture with economic incentives via a native token. While it achieves meaningful deployability, it shares Katzenpost's limitations regarding unverified mixing and client/relay separation, additionally introduces centralisation risks through token economics, does not support hidden services, and has not yet deployed post-quantum cryptography.

3.5 Differentiation Matrix

Property	Tor	I2P	Katzenpost	Nym	Phantom
GPA Resistance	✗	✗	Partial	Partial	☑
Verifiable Mix (ZK)	✗	✗	✗	✗	☑
Trustless PKI	✗	☑	✗	✗	☑
Hidden Services	☑	☑	✗	✗	☑
True P2P (client=node)	✗	☑	✗	✗	☑
Post-Quantum (deployed)	✗	✗	☑	✗	☑
DC-Net Mode	✗	✗	✗	✗	☑
Cover Traffic	✗	✗	☑	☑	☑

Table 1 — Feature comparison across major anonymous communication protocols

4. Protocol Architecture

Phantom Protocol is structured as a layered stack. Each layer has a precisely defined responsibility and interface. The layers are, from application to physical:

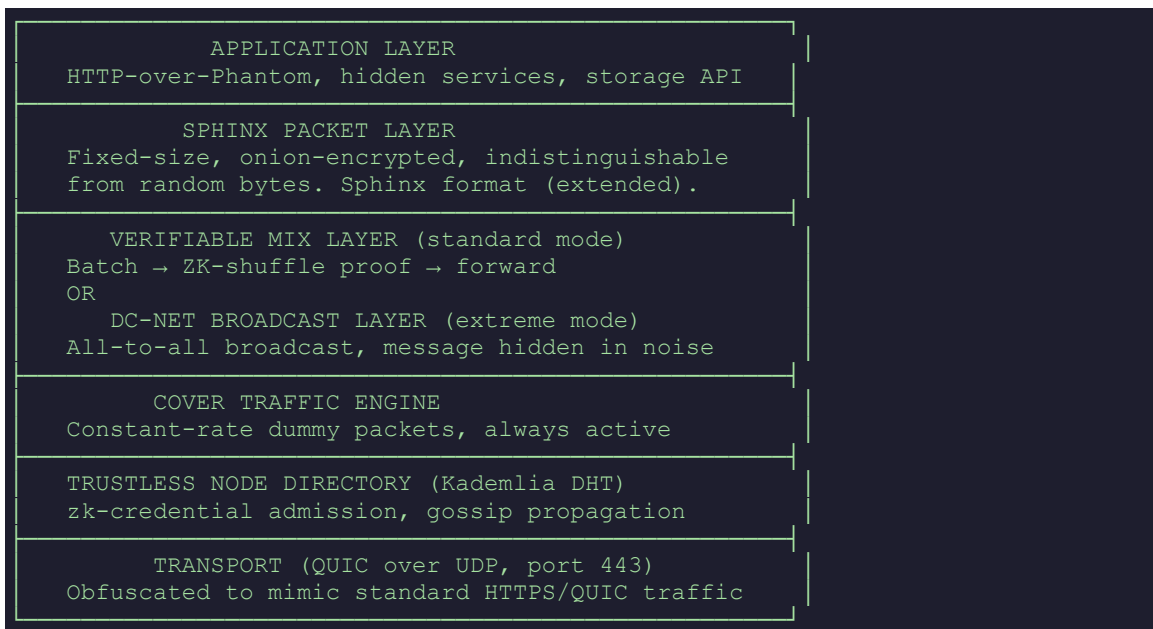


Figure 1 — Phantom Protocol layer stack

4.1 Node Roles

In Phantom Protocol, every participant is a node. There is no distinction between clients and relays. Every device running Phantom Protocol software simultaneously sends, receives, and routes traffic for others. This design ensures that the anonymity set is identical to the total user population — the largest possible value of k in the anonymity guarantee.

Nodes are categorised not by role but by capacity contribution: the fraction of their bandwidth and computation they contribute to the mix network. Low-capacity nodes (mobile devices) contribute minimally; high-capacity nodes contribute significantly. All nodes route traffic regardless of capacity tier.

4.2 Dual-Mode Operation

Phantom Protocol operates in two modes, selectable per-message:

- ▶ Standard Mode uses Sphinx packet routing through a stratified mix network with verifiable shuffling. Latency target: 200–800ms. Suitable for web browsing, file transfer, and general communication.
- ▶ Extreme Mode uses DC-Net broadcast for information-theoretically anonymous transmission. Latency: 5 seconds to several minutes depending on group size. Suitable for whistleblowing, high-risk communication, and cases where the sender's existence — not just identity — must be concealed.

5. Cryptographic Primitives

All cryptographic choices are made with the following constraints: security against both classical and quantum adversaries; independence from trusted setup (no ceremonies, no trapdoors); and composability within the Sphinx packet format and ZK proof systems.

5.1 Primitive Selection

Function	Classical	Post-Quantum	Mode
Key Encapsulation	X25519	CRYSTALS-Kyber-1024	Hybrid (both)
Digital Signatures	Ed25519	CRYSTALS-Dilithium-3	Hybrid (both)
Symmetric Cipher	ChaCha20-Poly1305	—	Unchanged (safe)
Hash Function	BLAKE3	—	Unchanged (safe)
ZK Proof System	Groth16	STARK (roadmap)	Groth16 now
Mix Proof	Bayer-Groth Shuffle	—	Standard mode
Node Identity	H(pubkey)	H(kyber_pubkey)	Self-certifying

Table 2 — Cryptographic primitive selection

5.2 Hybrid Key Exchange

Phantom Protocol uses a hybrid key exchange that combines X25519 (classical) and CRYSTALS-Kyber-1024 (post-quantum). Both are computed independently and their shared secrets are combined via BLAKE3:

```
shared_secret = BLAKE3(
    X25519(ephemeral_sk, node_pk_classical)
    || Kyber.Decapsulate(ephemeral_ct, node_sk_pq)
    || context_string
)
```

This hybrid construction ensures security if either primitive is broken — classical security holds against current adversaries, post-quantum security holds against future quantum adversaries.

5.3 Zero-Knowledge Shuffle Proofs

The verifiable mix layer requires each node to prove that its output batch is a permutation of its input batch — with each packet correctly decrypted — without revealing the permutation. Phantom Protocol uses the Bayer-Groth verifiable shuffle argument, which produces a constant-size proof regardless of batch size.

The proof is published to the DHT and verifiable by any network participant. Nodes that fail to produce a valid proof within the batch window are ejected from the node directory and flagged in the gossip layer.

```
For batch B = [p_1, p_2, ..., p_n]:
1. Node decrypts one Sphinx layer from each packet → B'
2. Node shuffles B' → B'' (permutation σ, kept secret)
3. Node computes π = BayerGroth.Prove(B', B'', σ)
4. Node publishes (B'', π) to next mix layer
5. Any verifier: BayerGroth.Verify(B', B'', π) → {valid, invalid}
```

6. Sphinx Packet Format (Extended)

Phantom Protocol uses an extended version of the Sphinx packet format. Every packet has an identical size (1452 bytes — one Ethernet MTU minus IP/UDP/QUIC overhead). All packets are indistinguishable from random bytes to any observer who does not hold the appropriate key material.

```
Phantom Sphinx Packet (1452 bytes):

HEADER (256 bytes)
├─ Blinded group element (X25519+Kyber): 96B
├─ Routing info (onion-encrypted): 128B
└─ Per-hop MAC (BLAKE3): 32B

BODY (1196 bytes)
└─ Payload (encrypted, PKCS7-padded)
```

The routing info encodes the next-hop address, encrypted under the current node's key. When a node processes a packet, it decrypts its layer, reads the next-hop address, and removes its layer — the resulting packet is identical in size to the input, as the routing info field is shifted and a random block appended.

Cover traffic packets are structurally identical to real packets. They contain encrypted dummy payloads that are discarded at the exit node (or looped back as loop cover). No node — including the exit — can distinguish a cover packet from a real packet.

7. Trustless Node Directory

The absence of directory authorities is the most structurally significant difference between Phantom Protocol and all existing mix networks. Phantom replaces centralised directories with a three-component trustless PKI.

7.1 Node Identity

A node's identity is derived entirely from its cryptographic keys — no registration with any authority is required:

```
node_id = BLAKE3(ed25519_pubkey || dilithium_pubkey || kyber_pubkey)
```

This identity is self-certifying: any participant can verify that a node's claimed identity matches its keys without consulting any third party.

7.2 Sybil Resistance via Proof-of-Work

To join the node directory, a new node must solve a moderately difficult PoW puzzle tied to its node_id and the current network epoch. The puzzle difficulty adjusts to target a network-wide node admission rate, preventing rapid Sybil attacks without requiring trusted registration.

```
admission_token = PoW(node_id || epoch || nonce)
where H(admission_token) < target_difficulty
```

7.3 DHT-Based Discovery

Nodes publish their descriptors (node_id, IP, port, public keys, capacity tier, PoW admission token, epoch signature) to a Kademlia DHT. Descriptors are signed by the node's Ed25519+Dilithium keys. Any node can verify any descriptor without contacting the publishing node.

Descriptors expire at the end of each epoch (default: 1 hour). Nodes must re-solve the PoW puzzle each epoch to remain in the directory. This limits the window of usefulness of a compromised descriptor and ensures the directory reflects currently-online nodes.

7.4 Gossip Propagation

The DHT is augmented with a gossip protocol for rapid propagation of node ejections (invalid shuffle proofs, equivocation, detected misbehaviour). Ejection notices are signed by the detecting node and propagated to all peers within seconds, preventing a misbehaving node from serving additional traffic while its DHT entry expires.

8. Hidden Services

Phantom Protocol provides native hidden service support — the ability to host a server whose IP address is permanently unknown to all clients, all relays, and all observers. This fills a critical gap left by both Katzenpost and Nym.

8.1 Address Derivation

A hidden service address is derived from the service's long-term public key:

```
service_address = BLAKE3(ed25519_pubkey || kyber_pubkey)[:32]
  represented as a 52-character Base32 string with .phantom TLD
```

```
Example: 3xk7m2nqvabcdefgghijklmnopqrstuvwxyz4567890aa.phantom
```

The address is self-authenticating: a client connecting to a .phantom address automatically verifies that the server holds the corresponding private key, preventing impersonation without any PKI.

8.2 Rendezvous Protocol

Client and server establish a connection through a rendezvous point inside the Phantom network — a randomly selected honest node that neither party knows the other's IP address through:

- ▶ Server selects N introduction nodes and registers with each via Sphinx-routed messages.
- ▶ Server publishes introduction node list to DHT, signed with service key.
- ▶ Client retrieves introduction node list, selects a rendezvous node, and sends a rendezvous request to the server via an introduction node.
- ▶ Server builds a Sphinx circuit to the rendezvous node. Client and server exchange keys through the rendezvous node and derive a shared session key.
- ▶ All subsequent communication is encrypted end-to-end and routed through the mix network. The rendezvous node sees only encrypted traffic.

9. DC-Net Extreme Mode

For the highest-risk communications — whistleblowers, journalists in hostile environments, dissidents under active surveillance — Phantom Protocol offers DC-Net mode, which provides information-theoretic sender anonymity. Unlike all other anonymity mechanisms, DC-Net anonymity does not depend on any computational hardness assumption. It is secure even against an adversary with unlimited computational power who observes all network traffic.

9.1 Operation Principle

DC-Net mode operates within a group of N participants. In each round, every participant broadcasts exactly one message of fixed size. The real message from the actual sender is XOR-combined with random pads shared with each other participant. All other participants broadcast only their random pads. The XOR of all broadcasts reveals the real message, but any subset of N-

1 participant is computationally and informationally indistinguishable from any permutation of senders.

```
For group G = {n_1, n_2, ..., n_N}:  
  
  Each pair (n_i, n_j) shares a random pad: pad_ij  
  Sender n_s wants to send message M  
  
  Broadcast from n_s: M XOR pad_s1 XOR pad_s2 XOR ... XOR pad_sN  
  Broadcast from n_k (k≠s): pad_k1 XOR pad_k2 XOR ... XOR pad_kN  
  
  XOR of all broadcasts = M (all pads cancel)  
  No participant or observer can determine which node sent M
```

9.2 Collision Resolution

When two participants attempt to send simultaneously, a collision occurs that is detectable (the XOR of both messages appears). Phantom Protocol uses a reservation protocol: before each round, participants use a lightweight coordination phase to claim a transmission slot. Slot claims are themselves anonymised through a smaller DC-Net, ensuring that even the act of claiming a slot reveals nothing about the claimant's identity.

9.3 Group Formation

DC-Net groups are formed dynamically from participants in a geographic or network proximity cluster. Group size is bounded between 8 and 128 participants — below 8, the anonymity set is too small; above 128, coordination overhead becomes excessive. Group membership is rotated every 10 minutes to prevent long-term intersection attacks.

10. Cover Traffic Engine

Cover traffic is mandatory and non-negotiable in Phantom Protocol. Every node sends exactly one packet per time slot (default: 100ms) regardless of whether it has real traffic to send. This eliminates the single most powerful attack against mix networks: timing correlation.

Three types of cover traffic are generated:

- ▶ Loop cover: packets routed through the mix network that return to the originating node, verifying that the node's circuits are functioning correctly.
- ▶ Drop cover: packets routed to a randomly selected exit node and discarded. These are indistinguishable from real traffic to all relay nodes.
- ▶ Decoy traffic: packets addressed to real nodes carrying encrypted dummy payloads. Recipients discard dummy payloads silently.

Cover traffic generation is handled by the Phantom Protocol daemon and is not configurable by the user. Users cannot disable cover traffic. This is an intentional design constraint — optional cover traffic provides no anonymity guarantee because its absence is itself a signal.

11. Implementation Roadmap

Phantom Protocol will be implemented in Rust, using the async runtime Tokio and the QUIC implementation Quinn. The following roadmap targets a production-ready v1.0 release within 18 months of project inception.

Phase	Timeline	Deliverable	Milestone
Phase 0	Months 1–2	Protocol specification complete	This whitepaper + formal spec
Phase 1	Months 2–5	Core primitives library	Sphinx impl, hybrid KEM, DHT
Phase 2	Months 5–9	Mix network + ZK proofs	Verifiable shuffle, private testnet
Phase 3	Months 9–12	Cover traffic + obfuscation	Pluggable transport, traffic shaping
Phase 4	Months 12–15	Hidden services + DC-Net	Rendezvous protocol, DC-Net groups
Phase 5	Months 15–18	Hardening + audit	Security audit, public testnet
Phase 6	Month 18+	Ecosystem + governance	Client libs (Python, JS), open governance

Table 3 — Implementation roadmap

11.1 Reference Implementation Stack

Component	Library / Crate	Purpose
Async runtime	tokio	Async I/O foundation
QUIC transport	quinn	Obfuscated UDP transport
DHT	libp2p (Kademlia)	Trustless node directory
Gossip	libp2p (GossipSub)	Node ejection propagation
Sphinx packets	Custom (Rust)	Packet format implementation
ZK proofs	arkworks-rs	Groth16 shuffle proofs
Post-quantum	pqcrypto	Kyber + Dilithium
Classical crypto	dalek-cryptography	X25519 + Ed25519
Hash	blake3	All hashing operations
PoW	Custom (Rust)	Sybil resistance

Table 4 — Reference implementation stack

12. Open Problems and Future Work

Phantom Protocol is honest about the problems it has not yet solved. The following areas represent active research questions that will be addressed iteratively:

- ▶ **Optimal tuning:** the mixing delay distribution and batch size parameters that minimise metadata leakage for a given anonymity set size and latency target remain an open research problem, consistent with the state of the art in mix network research.
- ▶ **DC-Net scalability:** the current DC-Net design is bounded at approximately 128 participants. Research into hierarchical DC-Nets or hybrid DC-Net/Mix architectures for larger groups is ongoing.
- ▶ **Mobile and constrained devices:** mandatory cover traffic creates a battery and bandwidth burden on mobile devices. A capacity-tiered model that allows mobile nodes to contribute asymmetrically while maintaining the anonymity guarantee requires further design work.
- ▶ **STARK migration:** Groth16 proofs require a trusted setup ceremony. Migrating to STARK-based proofs (which require no trusted setup) is on the roadmap but awaits maturation of Rust STARK libraries.
- ▶ **Incentive design:** unlike Nym, Phantom Protocol currently has no economic incentive for node operators. Designing an incentive mechanism that does not introduce centralisation, does not require a token, and is not gameable is an important open problem.

13. Security Considerations

⚠ Cryptographic Review Required

This is a draft specification. No component of Phantom Protocol should be used in production without independent cryptographic review by qualified professionals. The authors make no security guarantees for this draft.

The following security properties are claimed for a correctly implemented and deployed Phantom Protocol network:

- ▶ **Traffic analysis resistance:** mandatory constant-rate cover traffic eliminates the primary attack vector against mix networks. A GPA observing all links cannot correlate sender and receiver based on traffic patterns alone.
- ▶ **Compromise resilience:** the ZK shuffle proof ensures that even a fully compromised mix node cannot deanonymize traffic without detection. A compromised node that forwards packets in input order will be detected and ejected within one batch window.
- ▶ **Post-quantum security:** the hybrid key exchange ensures security against quantum adversaries. All long-term keys use post-quantum algorithms; all ephemeral keys use both classical and post-quantum constructions.
- ▶ **No trusted setup requirement (roadmap):** the current Groth16 ZK system requires a trusted setup ceremony. The STARK migration will eliminate this requirement entirely, removing the last trusted party from the system.

14. Governance and Openness

Phantom Protocol is an open protocol. No entity — including its original authors — has exclusive authority over the protocol specification. Governance follows the model established by successful open protocol communities:

- ▶ The protocol specification is maintained as an open document with versioned releases. Changes require consensus among active implementors.
- ▶ The reference implementation is released under a permissive open-source license (Apache 2.0 or MIT, to be determined).
- ▶ No foundation, company, or individual holds trademark or patent claims over the protocol or its name.
- ▶ Any individual or organisation may implement the protocol, operate nodes, or build applications without permission or fees.

The protocol's openness is not merely philosophical — it is a security property. A protocol controlled by any identifiable entity can be compelled. A truly open protocol with multiple independent implementations has no centre to attack.

15. References

The following works are foundational to Phantom Protocol's design. Readers are encouraged to study them in the order listed.

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- ▶ Groth, J. (2016). On the Size of Pairing-Based Non-interactive Arguments. *EUROCRYPT 2016*. (Groth16 ZK system)

Appendix A — Notation

Symbol	Meaning
GPA	Global Passive Adversary
k	Anonymity set size
N	Number of nodes or DC-Net group members
π	Zero-knowledge proof
σ	Shuffle permutation (secret)
B, B', B''	Packet batch (input, decrypted, shuffled)
$H(\cdot)$	BLAKE3 hash function
KEM	Key Encapsulation Mechanism
NIKE	Non-Interactive Key Exchange
MTU	Maximum Transmission Unit (1452 bytes for Phantom)
DHT	Distributed Hash Table (Kademlia)
PoW	Proof of Work
TLD	.phantom (Phantom Protocol Top-Level Domain)

Table A1 — Notation reference

— End of Phantom Protocol Whitepaper v0.1 —